

Answer: B

27 Total number of electrons in the cylinder = volume $\times$ no. of electron per unit volume
= area $\times$ length $\times$ no. of electron per unit volume
= $(3.2 \times 10^{-7}) \times (0.0047) \times (8.5 \times 10^{28})$
= 1.278×10^{20}

$$\text{Total current} = \frac{(1.278 \times 10^{20}) \times (1.6 \times 10^{-19})}{60} = 0.34 \text{ A}$$

Answer: A

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